Ryan Lee
Data Analytics
5 December 2025

Lab 6

The summary of a linear regression model:

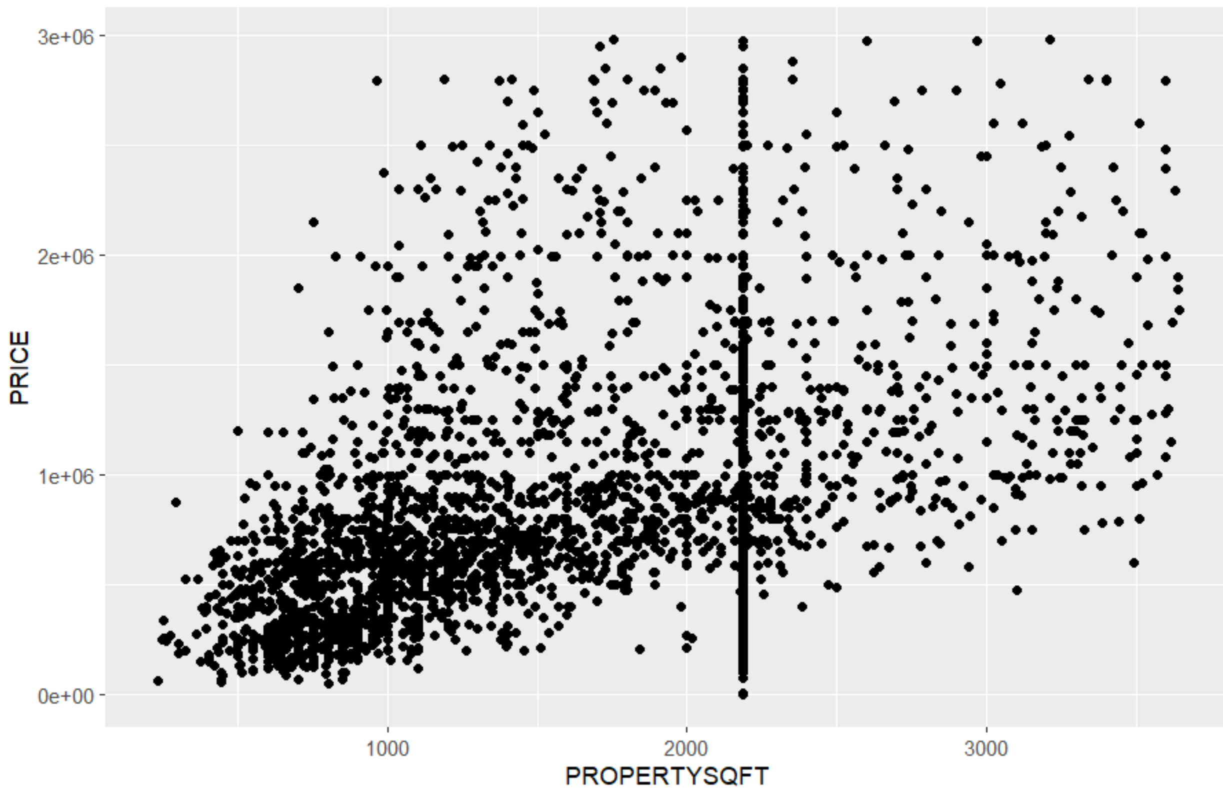
```
> summary(lin.mod1)

Call:
lm(formula = PRICE ~ PROPERTYSQFT, data = train)

Residuals:
    Min      1Q  Median      3Q     Max 
-998894 -323019 -103990  202481 2168535 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  329715.11   25775.41   12.79   <2e-16 ***
PROPERTYSQFT    307.51      13.87   22.17   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 534600 on 3065 degrees of freedom
Multiple R-squared:  0.1382,	Adjusted R-squared:  0.1379 
F-statistic: 491.4 on 1 and 3065 DF,  p-value: < 2.2e-16
```



This is results of a 10-fold Monte Carlo cross validation function:

```
10-Fold Cross Validation
method: Woodbury
criterion: mse
cross-validation criterion = 284395922096
bias-adjusted cross-validation criterion = 284380072695
95% CI for bias-adjusted CV criterion = (264348093576, 304412051814)
full-sample criterion = 284094748548
```

This is the evaluation metrics for the linear regression model:

```
> mean.abs.err
[1] 401303.5
> mean.sq.err
[1] 289811806440
> root.mean.sq.err
[1] 538341.7
```

The summary of the results from a SVM regression model:

```
> svm.mod

Call:
```

```
svm(formula = PRICE ~ PROPERTYSQFT, data = train, kernel = "linear", gamma =
gamma.range,
    cost = C.range)


Parameters:
   SVM-Type:  eps-regression
 SVM-Kernel:  linear
       cost:  1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
      gamma:  0.03125 0.0625 0.125 0.25 0.5 1 2 4
    epsilon:  0.1


Number of Support Vectors:  2514
```

The evaluation results of the SVM regression model:

```
> results1
       mae         mse    rmse
1 62073.84 47653905998 89057.3
```

This is the evaluation metrics of the random forest regression model:

```
> mean.abs.err
[1] 389029.3
> mean.sq.err
[1] 293162607183
> root.mean.sq.err
[1] 541444.9
```